

Statistical philosophy

Jonathan Dushoff , McMaster University

Bio 708 QMEE
Jan 2026

GOALS

- ▶ Discuss what statistics are used for, and why they are needed
- ▶ Explain what P values mean, and what they don't
 - ▶ Effect sizes and confidence intervals are usually better
- ▶ Explain the fundamentals of the two basic paradigms of statistical philosophy
- ▶ Discuss the role of statistics in science

Outline

Statistical inference

- P values and confidence intervals

- Statistics and science

Paradigms for inference

- Frequentist paradigm

- Bayesian paradigm

Philosophy of fitting

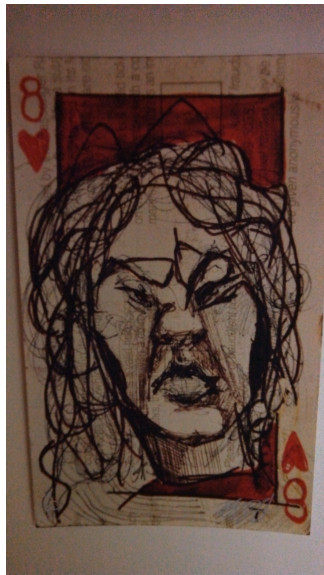
Conclusion

Statistical inference

- ▶ We use statistics to confirm effects, estimate parameters, and predict outcomes
- ▶ It usually rains when I'm in Cape Town, but mostly on Sunday
 - ▶ *Confirmation:* In Cape Town, it rains more on Sundays than other days
 - ▶ *Estimation:* In Cape Town, the *odds* of rain on Sunday are 1.6–2.2 times higher than on other days
 - ▶ *Prediction:* I am confident that it will rain at least one Sunday the next time I go

Raining in Cape Town

- ▶ How we interpret data like this necessarily depends on assumptions:
 - ▶ Is it likely our observations occurred by chance?
 - ▶ Is it likely they *didn't*?



Tessa Wessels, Faces on a Train

Vitamin A

- ▶ We compare health indicators of children treated or not treated with vitamin A supplements
 - ▶ *Estimate*: how much taller (or shorter) are the treated children on average?
 - ▶ *Confirmation*: are we sure that the supplements are helping (or hurting)?
 - ▶ *Range of estimates*: how much do we think the supplement is helping?

Outline

Statistical inference

- P values and confidence intervals

- Statistics and science

Paradigms for inference

- Frequentist paradigm

- Bayesian paradigm

Philosophy of fitting

Conclusion

P values and confidence intervals

- ▶ We use *P values* to say how sure we are that we have seen a positive effect
- ▶ We use *confidence intervals* to say what we think is going on (with a certain level of confidence)
- ▶ P values are *over-rated*
- ▶ *Never* use a high P value as (direct) evidence for anything, e.g.:
 - ▶ that an effect is small
 - ▶ that two quantities are similar
- ▶ What does a P value actually measure?

Vitamin A example

- ▶ We want to know if vitamin A supplements improve the health of village children
 - ▶ Is height is a good measure of general health?
 - ▶ How will we know height differences are due to our treatment?
 - ▶ *
 - ▶ *
 - ▶ *

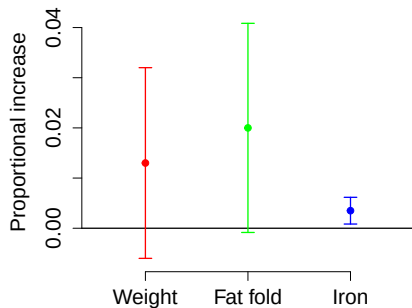
What do we hope to learn?

- ▶ Is vitamin A good for these children?
- ▶ How sure are we?
- ▶ How good do we think it is?
- ▶ How sure are we about that?

P values

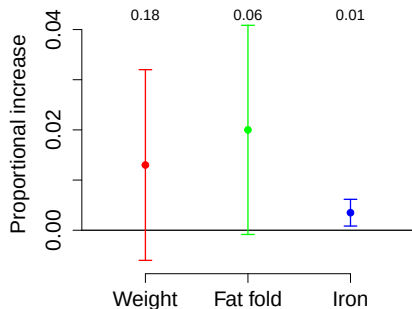
- ▶ Activity: What is a P value?
- ▶ What does it mean if I find a “significant P value” for some effect in this experiment?
- ▶ *
- ▶ *
- ▶ If I'm certain that the true answer isn't exactly zero, why do I want the P value anyway?

Confidence intervals



- ▶ What do these results mean?
- ▶ Which are significant?

Confidence intervals and P values



- ▶ A high P value means we can't see the *sign* of the effect clearly
- ▶ A low P value means we can
- ▶ What would happen if we could measure many more children?

What do P values measure?



► *

► *

► *

Types of Error

- ▶ Type I (*False positive:*) concluding there is an effect when there isn't one
 - ▶ This doesn't happen in biology. There is always an effect.
- ▶ Type II (*False negative:*) concluding there is no effect when there really is
 - ▶ This *should* never happen, because we should never conclude there is no effect

Types of Error

- ▶ Type Zero is the error of using numerical codes for things that have perfectly good simple names
- ▶ Just say “false positive” or “false negative” when possible

Experimental design

- ▶ *False positive*: in the *hypothetical* case that the effect is exactly zero, what is the probability of falsely finding an effect?
 - ▶ Should be less than or equal to my significance value
- ▶ *False negative*: what is the probability of failing to find an effect that is there?
 - ▶ Requires you specify a hypothetical effect *size*
 - ▶ This is a *scientific* judgment
- ▶ These are useful to analyze **power** and **validity** of a statistical design
 - ▶ You should do these analyses *before* you collect data

Doing Science

- ▶ Once you collect data:
 - ▶ You should never make a false-negative error.
 - ▶ *
 - ▶ Or a false-positive error
 - ▶ *
- ▶ *Sign error*: if I think an effect is positive, when it's really negative (or vice versa)
- ▶ *Magnitude error*: if I think an effect is small, when it's really large (or vice versa)
- ▶ Confidence intervals aid clear thinking

Low P values

- ▶ If I have a low P value I can see something clearly
- ▶ But it's usually better to focus on what I see than the P value



High P values

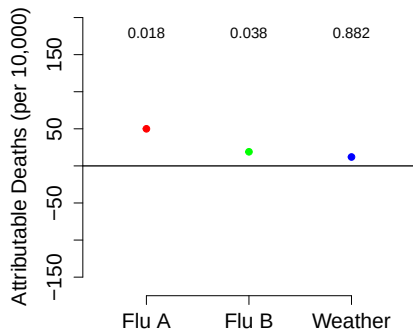
- ▶ If I have a high P value, there is something I *don't* see clearly
- ▶ It *may be* because this effect is small
- ▶ High P values should *not* be used to advance *any* conclusion



What causes high P values?

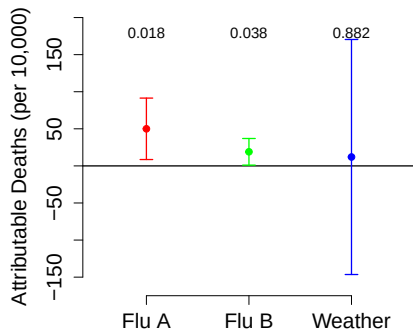
- ▶ Small differences
- ▶ Less data
- ▶ More noise
- ▶ An inappropriate model
- ▶ Less model resolution
- ▶ A lower P value means that your evidence for difference is better
- ▶ A higher P value means that your evidence for similarity is better – or worse!

Annualized flu deaths



- ▶ Look at deaths on an annual time scale
- ▶ Why is weather not causing deaths at this time scale?
- ▶ *

... with confidence intervals



- ▶ **Never** say: A is significant and B isn't, so $A > B$
- ▶ **Instead:** Construct a statistic for the hypothesis $A > B$
 - ▶ May be difficult

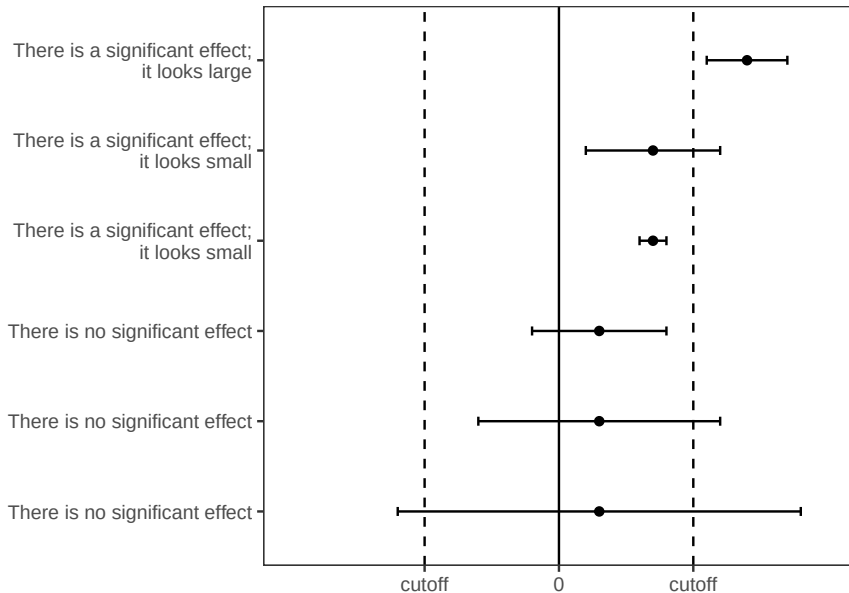
Misleading language

- ▶ Null effects of boot camps
 - ▶ *
- ▶ Fat storage in vole populations is not correlated with elevation
 - ▶ *
- ▶ As expected, the placebo group did not differ significantly from the control group
 - ▶ *
- ▶ B and B showed that there is no statistically significant difference in sexual risk behaviour between men with and without clinic access in Zambia
 - ▶ *
 - ▶ *

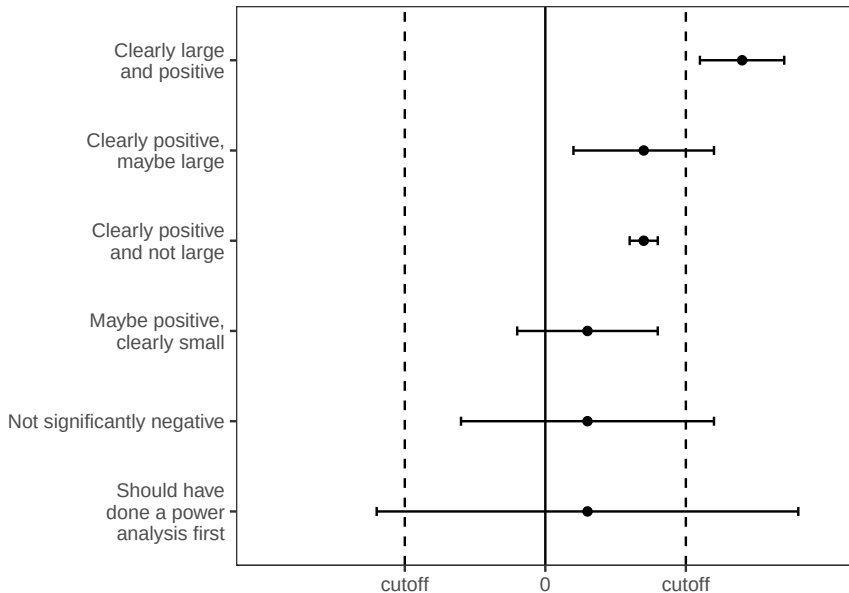
Another way to talk

- ▶ Unclear effects of boot camps
 - ▶ *
- ▶ As expected, the sign of the difference between the placebo group and controls was unclear
 - ▶ *
- ▶ The direction of correlation between fat storage and elevation in vole populations is unclear in this study
- ▶ B and B showed that there is an unclear difference in sexual risk behaviour between men with and without clinic access in Zambia
 - ▶ *

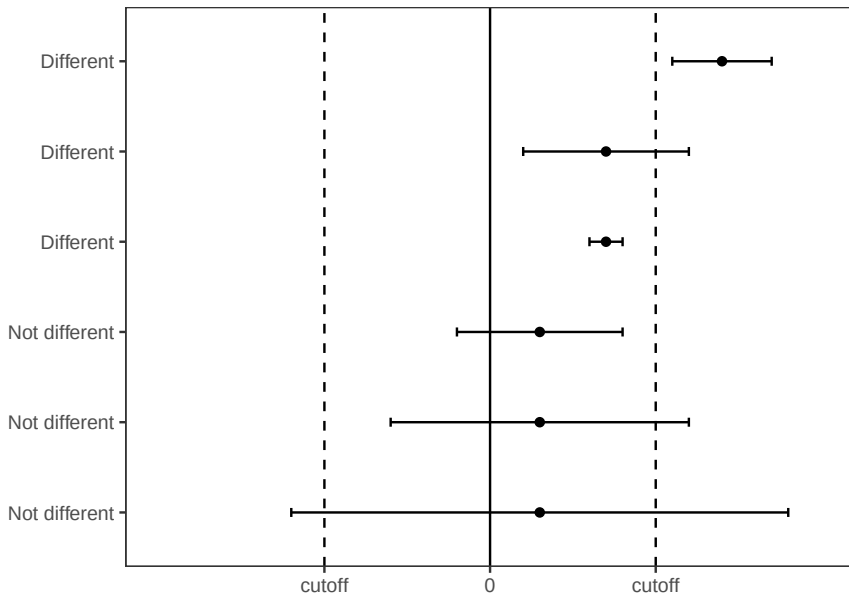
Confidence intervals



Confidence intervals



Confidence intervals



Cutoffs

- ▶ How do I know what is a large effect?
- ▶ This can't be done by statistics!
 - ▶ Must be scientific
 - ▶ It's often hard, and people avoid it
 - ▶ This leads to sloppy thinking

Outline

Statistical inference

P values and confidence intervals

Statistics and science

Paradigms for inference

Frequentist paradigm

Bayesian paradigm

Philosophy of fitting

Conclusion

Syllogisms

- ▶ All men are mortal
- ▶ Mohamed Salah is mortal
- ▶ Therefore, Mohamed Salah is a man



Syllogisms

- ▶ All men are mortal
- ▶ Fanny the elephant is mortal
- ▶ Therefore, Fanny the elephant is a man



Bad logic

- ▶ A lot of statistical practice works this way:
 - ▶ bad logic in service of conclusions that are (usually) correct
- ▶ This sort of statistical practice leads in the aggregate to bad science
- ▶ The logic can be fixed:
 - ▶ Estimate a difference, or an interaction

Small effects

- ▶ We can't build statistical confidence that something is small by failing to see it clearly
- ▶ We must instead see clearly that it is small
- ▶ This means we need a standard (our cutoff) for what we mean by small

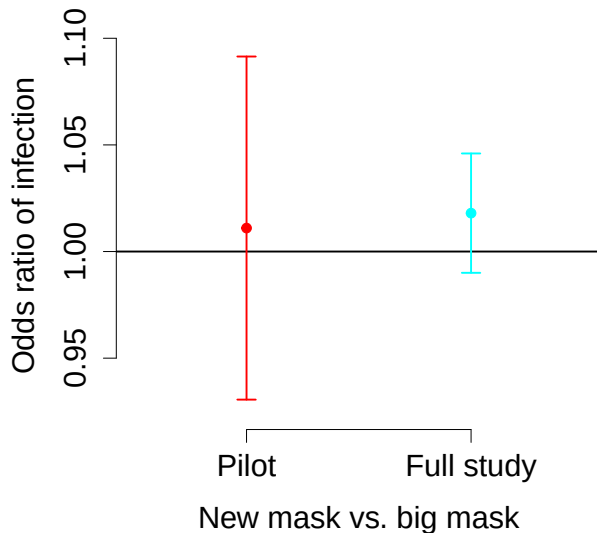
Flu masks



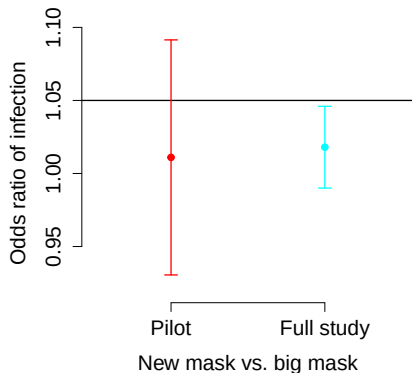
Flu mask example

- ▶ People who work in respiratory clinics sometimes have to wear bulky, uncomfortable, expensive masks
- ▶ They would like to switch to simpler masks, if those will do the job
- ▶ How can this be tested statistically? We don't want the masks to be “different”.
 - ▶ We need to decide what we mean by different in this case!
 - ▶ They're not the same, so how close is close enough?

Traditional statistics

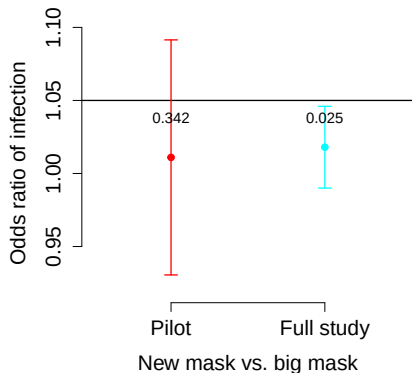


Non-inferiority approach



- ▶ Are we confident the new mask is “good enough”?
- ▶ There is no substitute for picking a cutoff
- ▶ Can be controversial

Non-inferiority approach



- ▶ We can even attach a P value by basing it on the “right” statistic.
- ▶ The right statistic is the thing whose sign we want to know:
 - ▶ The difference between the observed effect and the standard we chose

Outline

Statistical inference

P values and confidence intervals

Statistics and science

Paradigms for inference

Frequentist paradigm

Bayesian paradigm

Philosophy of fitting

Conclusion

Outline

Statistical inference

P values and confidence intervals

Statistics and science

Paradigms for inference

Frequentist paradigm

Bayesian paradigm

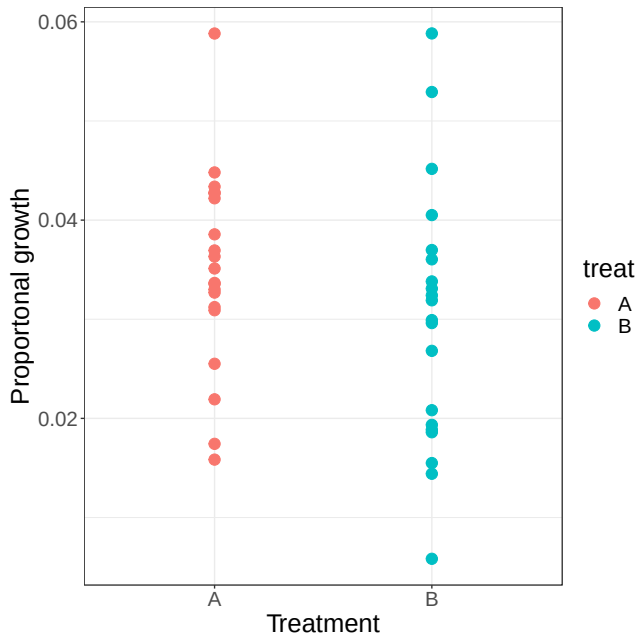
Philosophy of fitting

Conclusion

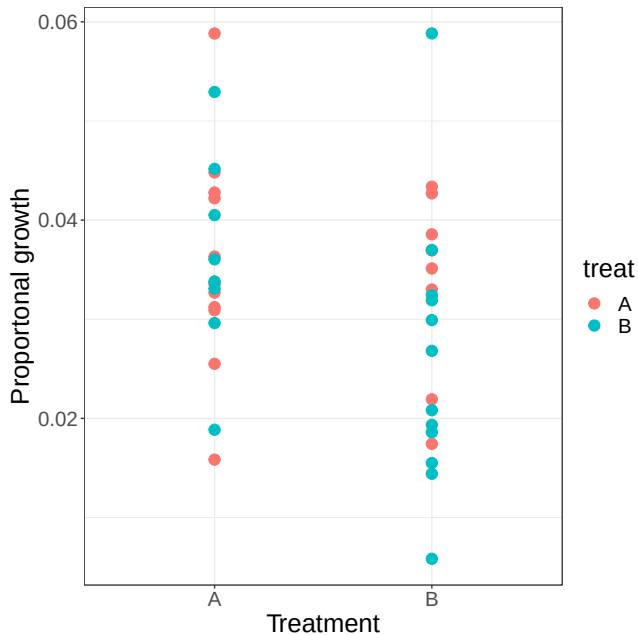
Frequentist paradigm

- ▶ Make a null model
- ▶ Test whether the effect you see could be due to chance
 - ▶ What is the probability of seeing a difference of exactly 0.0048 in proportional growth?
 - ▶ *
- ▶ Test whether the effect you see *or a larger effect* could be due to chance
 - ▶ This probability is the (frequentist) P value
- ▶ JD: It's actually better to calculate a one-sided P value. You would then usually double this to get a standard P value

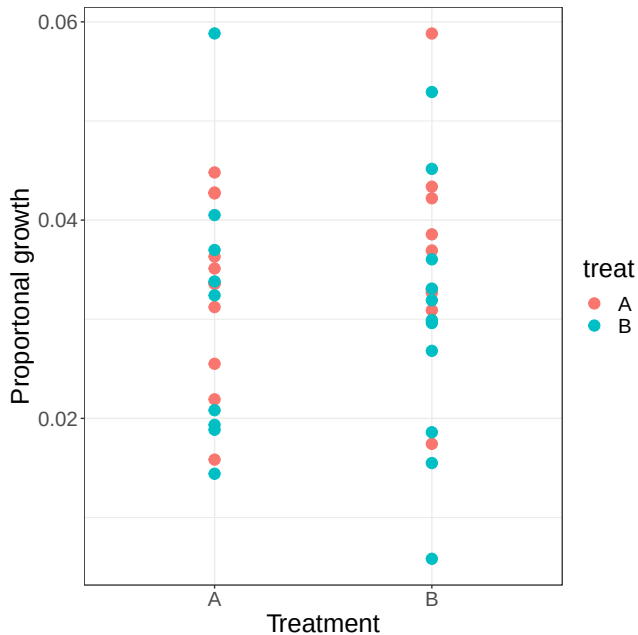
Height measurements



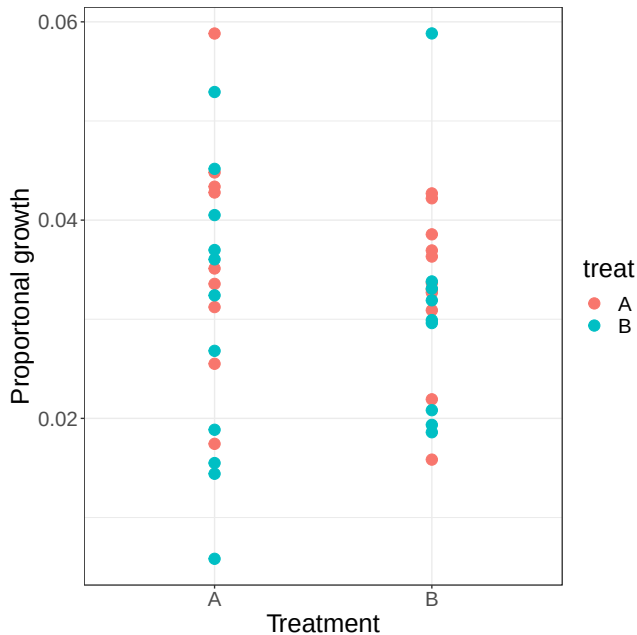
Scrambled measurements



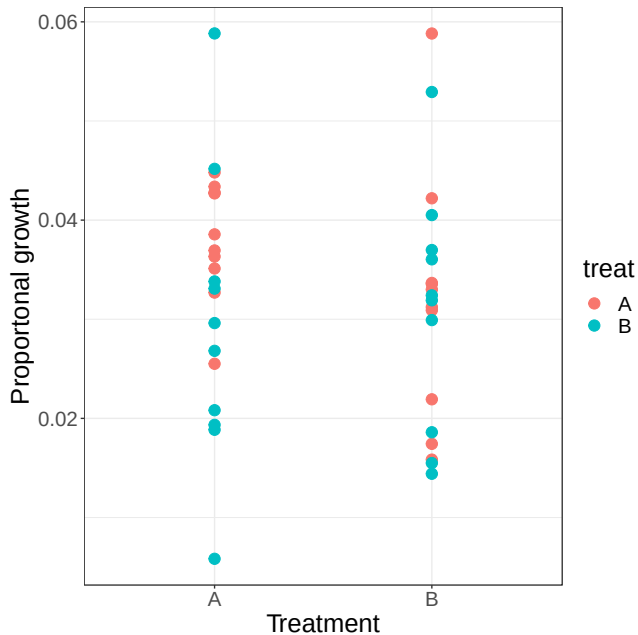
Scrambled measurements



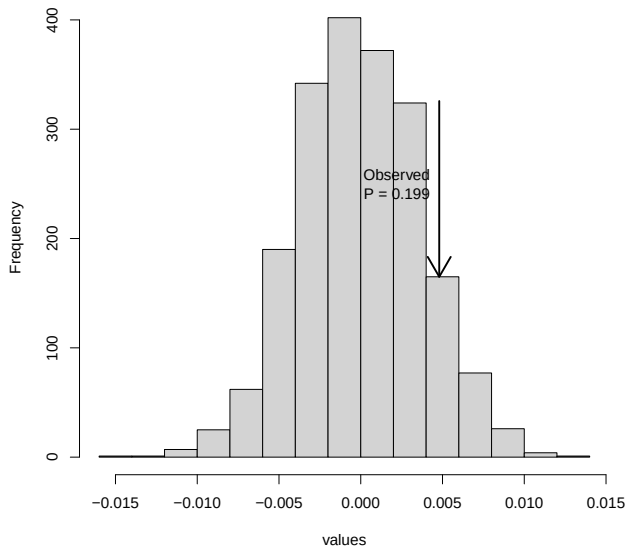
Scrambled measurements



Scrambled measurements



The null distribution



Outline

Statistical inference

- P values and confidence intervals

- Statistics and science

Paradigms for inference

- Frequentist paradigm

- Bayesian paradigm

Philosophy of fitting

Conclusion

Bayesian paradigm

- ▶ Make a complete model world
- ▶ Use conditional probability to calculate the probability you want



A powerful framework

- ▶ More assumptions $\implies$ more power
- ▶ With great power comes great responsibility



Bayesian inference

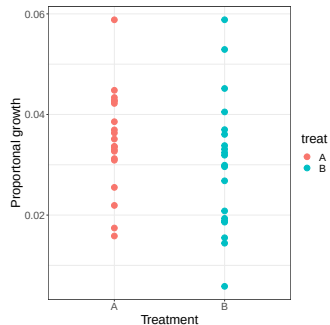
- ▶ We want to go from a *statistical model* of how our data are generated, to a probability model of parameter values
 - ▶ Requires *prior* distributions:
 - ▶ the assumed likelihood of parameters before these observations are made
 - ▶ Use Bayes theorem to calculate *posterior* distributions:
 - ▶ the inferred likelihood of parameters after taking the data into account
- ▶ Provides a strong framework for combining information from different sources and for propagating uncertainty

Bayesian P value

- ▶ Once you make strong-enough assumptions, the Bayesian framework allows you to calculate true probabilities
 - ▶ Meaning, they are really probabilities; they are only as “true” as your assumptions
- ▶ A natural approach to Bayesian P values is simply the probability that your estimate has the wrong sign
 - ▶ Often better to double this value, to align with frequentist tradition

Vitamin A study

- ▶ A frequentist can do a clear analysis right away
- ▶ A Bayesian needs a ton of assumptions – will often try to make “uninformative” assumptions



Example: MMEV

- ▶ MMEV is a viral infection that can cause a serious disease (called MMED)
- ▶ MMED patients are unable to control their urge to fit models to data
- ▶ The rapid MMEV test gives a positive result:
 - ▶ 100% of the time for people with the virus
 - ▶ 5% of the time for people without the virus

MMED MMEV questions

- ▶ The rapid MMEV test gives a positive result:
 - ▶ 100% of the time for people with the virus
 - ▶ 5% of the time for people without the virus
- ▶ The population prevalence of MMEV is 1%
- ▶ You test a *random* person from this population, and the result is positive.
 - ▶ What is the probability that they have MMEV?
- ▶ This calculation is the core of Bayes *theorem*

MMED MMEV questions

- ▶ You learn that your friend has had a positive rapid test for MMEV
 - ▶ What do you tell them?
 - ▶ *
- ▶ Bayesian inference:
 - ▶ Clear conclusions
 - ▶ based on clear (and often strong) assumptions
 - ▶ Need to be clear about the assumptions you are making
- ▶ Medical “paradox” video

Outline

Statistical inference

- P values and confidence intervals

- Statistics and science

Paradigms for inference

- Frequentist paradigm

- Bayesian paradigm

Philosophy of fitting

Conclusion

Philosophy of fitting

- ▶ Principled statistical fitting is hard
- ▶ A big advantage of having good computers is that you can *test* fitting approaches on simple and complicated model worlds before applying them to real data

Approximating probabilities

- ▶ In both paradigms, we want to calculate true probabilities to reinforce our conclusions
- ▶ In most complicated situations (e.g., dynamical model fits), these probabilities cannot be calculated directly
- ▶ We approximate the probabilities we want using likelihoods

Likelihoods

- ▶ What we want to know is [frequentist or Bayesian] probabilities relating to our *parameters* based on the data
 - ▶ Is $\mathcal{R}_0 < 2$; Is $\bar{D} < 6d$?
- ▶ What we can easily calculate based on assumptions is the likelihood of our *data* based on the parameters.
 - ▶ Proportional to probability of seeing the data
 - ▶ But probability of seeing these exact data is hard to define, and not very interesting

Algorithms

- ▶ If we can calculate likelihoods, we have good algorithms to estimate the probabilities that we want
 - ▶ For frequentist tests, we use maximum likelihood estimation
 - ▶ For Bayesian tests, we use MCMC

Normal likelihoods

- ▶ Assume that the difference between the estimate $\hat{y}_i$ and the data point y_i is normally distributed. What is the log likelihood?



$$L = \prod_i \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$



$$\ell = \sum_i -\log(\sigma\sqrt{2\pi}) - \sum_i \frac{(y_i - \hat{y}_i)^2}{2\sigma^2}$$

- ▶ *We maximize the likelihood by minimizing the sum of squares*
 - ▶ and then solving for σ

Least squares $\rightarrow$ likelihood

- ▶ Attaching your least squares fit to a likelihood means:
 - ▶ You can *use it* for statistical inference (LRT)
 - ▶ You can *challenge* the assumptions

Other distributions

- ▶ Maybe we want a lognormal distribution of titres
 - ▶ Dr. Rong did that: least squares on the log scale!
- ▶ Maybe we want a Poisson (or negative binomial) distribution for counts
- ▶ Maybe we want a binomial (or beta binomial) distribution for samples
- ▶ It's better to be explicit about what distribution you are assuming!

Outline

Statistical inference

- P values and confidence intervals

- Statistics and science

Paradigms for inference

- Frequentist paradigm

- Bayesian paradigm

Philosophy of fitting

Conclusion

Your philosophy

- ▶ Statistics are not a magic machine that gives you the right answer
- ▶ If you are to be a serious scientist in a noisy world, you should have your own philosophy of statistics
 - ▶ Be pragmatic: your goal is to do science, not get caught by theoretical considerations
 - ▶ Be honest: it's harder than it sounds.

Honesty

- ▶ You can always keep analyzing until you find a “significant” result
 - ▶ If you do this you will make a lot of mistakes
- ▶ You may also keep analyzing until you find a result that you already believe is true.
 - ▶ This is *confirmation bias*; you're probably right, but your project is not advancing science
- ▶ Good practice
 - ▶ Make an analysis plan before you start
 - ▶ If flexibility is allowed in your context:
 - ▶ keep an analysis journal as you go

Summary



Summary

- ▶ P values are over-rated
 - ▶ and often misunderstand
- ▶ High P values should not be used as evidence for anything ever.
 - ▶ They can provide indirect evidence. Wonderful. Find the direct evidence and use that instead.
- ▶ Effect sizes and confidence intervals are almost always better
- ▶ Use the right P value (or confidence interval) for your question
 - ▶ Non-inferiority tests, interactions

Summary

- ▶ Frequentist statistics makes weak assumptions, and finds logically weak formal conclusions:
 - ▶ These parameters are unlikely to produce a statistic this extreme by chance
- ▶ Bayesian statistics makes strong assumptions:
 - ▶ prior distributions must be fully specified
- ▶ ...and finds logically strong formal conclusions:
 - ▶ The probability that the effect value is in this range is X
 - ▶ These strong conclusions can be used directly for prediction with uncertainty

Summary

- ▶ Statistics are a key component of data-based science
 - ▶ You should think about statistical analysis from the beginning of your project
- ▶ You need a basic understanding of statistical principles
- ▶ You need your own statistical philosophy
 - ▶ If you're a theoretician, it should be ideological and honest
 - ▶ If you're a scientist, it should be pragmatic and honest